

Introduction

Wireless power transfer (WPT) is currently found in the charging devices of everyday items, such as smartwatches, toothbrushes, and phones. **Wireless resonant inductive coupling** is a form of WPT where transmitter and receiver coils are tuned to the same resonant frequency, enhancing **power transfer efficiency (PTE)** [1]. This form of power transfer can be used in **dynamic systems** to charge EVs **more efficiently**. However, a major obstacle remains: PTE decreases



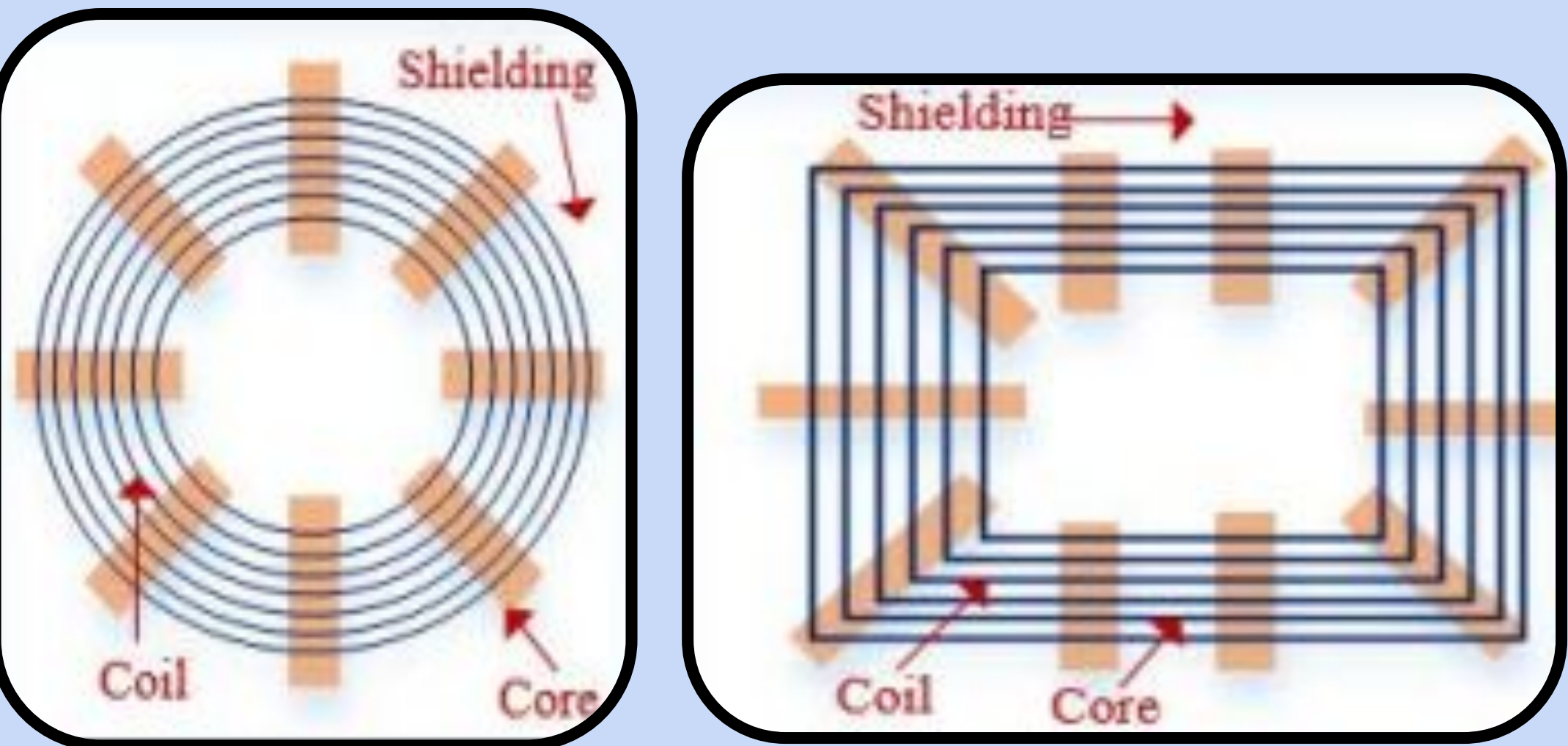
Figure 1: Render of an early dynamic WPT EV charging design. [2]

during **lateral and longitudinal misalignments** between coils and varies depending on the **shape of the coil** used [1][3].

Purpose

Research Question: How do various coil structures and misalignment conditions affect the power transfer efficiency of a dynamic wireless resonant inductive power transfer (WRIPT) system?

Hypothesis: The rectangular coil will maintain more efficient power transfer than the other coils shapes because its longer shape and sharper edges allow it to be less susceptible to power losses during misalignment conditions.



Figures 2 and 3: Diagrams displaying two different coil shapes—circular (left) and rectangular (right). [1]

Finding Optimal Coil Shape for Use in Wireless Resonant Inductive Power Transfer EV Charging Systems

Methods

Independent Variables

- Coil shape
- Misalignment condition (latitudinal/longitudinal) as a % of coil width

Dependent Variable

- PTE

Controls

- 0% misalignment tests for each coil shape

Procedure

1. Place **transmitter coils** and the **receiver coil** on separate **acrylic sheets**.
2. Connect **transmitter coils** in series to **function generator/oscilloscope** with a **capacitor** and **resistor**.
3. Connect the **receiver coil** to a **capacitor, resistor, and multimeter** in series.
4. Record initial **input voltage** with the **multimeter**.
5. Place **receiver circuit** directly above each **transmitter coil** and record voltage individually for **control tests**.
6. Shift **transmitter coil longitudinally/latitudinally** and record voltage individually for **misalignment tests**.
7. Complete **5 trials** for each **misalignment %** and **coil shape**.
8. Calculate **input/output power** with **voltage** and **resistance**.
9. Calculate **PTE** with $P_{in}/P_{out} * 100\%$.

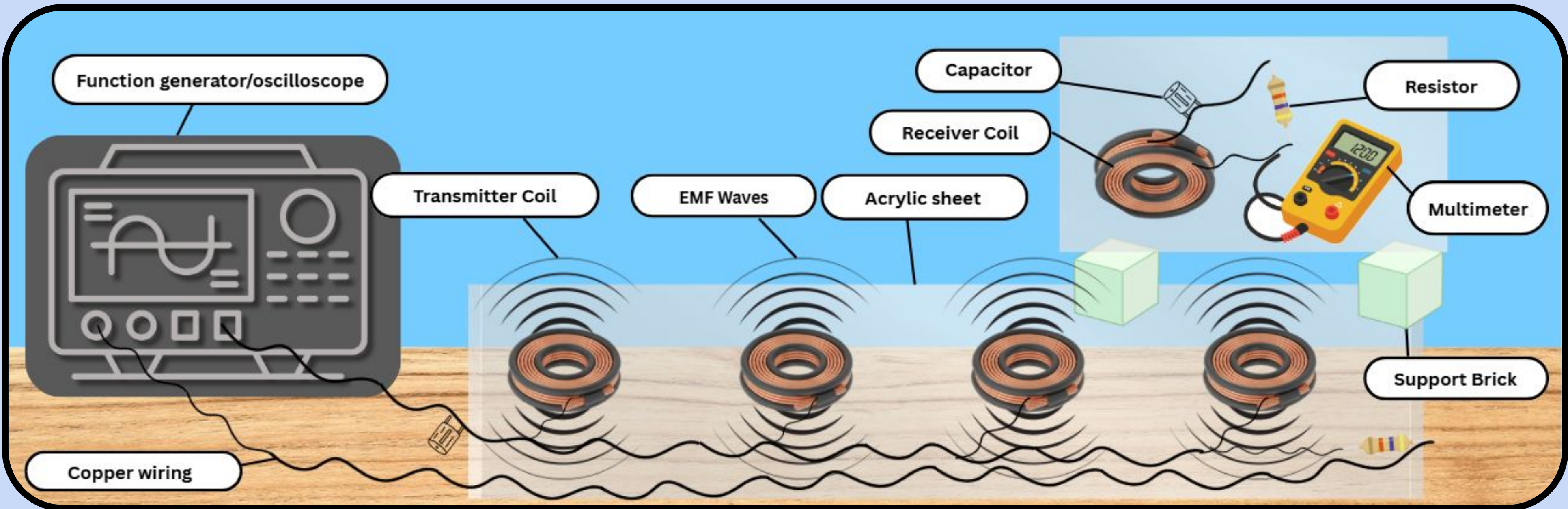


Figure 4: Illustration of the procedural setup. Image created by researcher using Canva, 2026.

***Note:** This diagram represents the setup for one trial.

Results

- Figure 5 shows that PTE generally **decreased as coil misalignment increased** across all coil geometries.
- Circular coils produced the **highest recorded values**.
- Efficiency changes were **non-linear**.

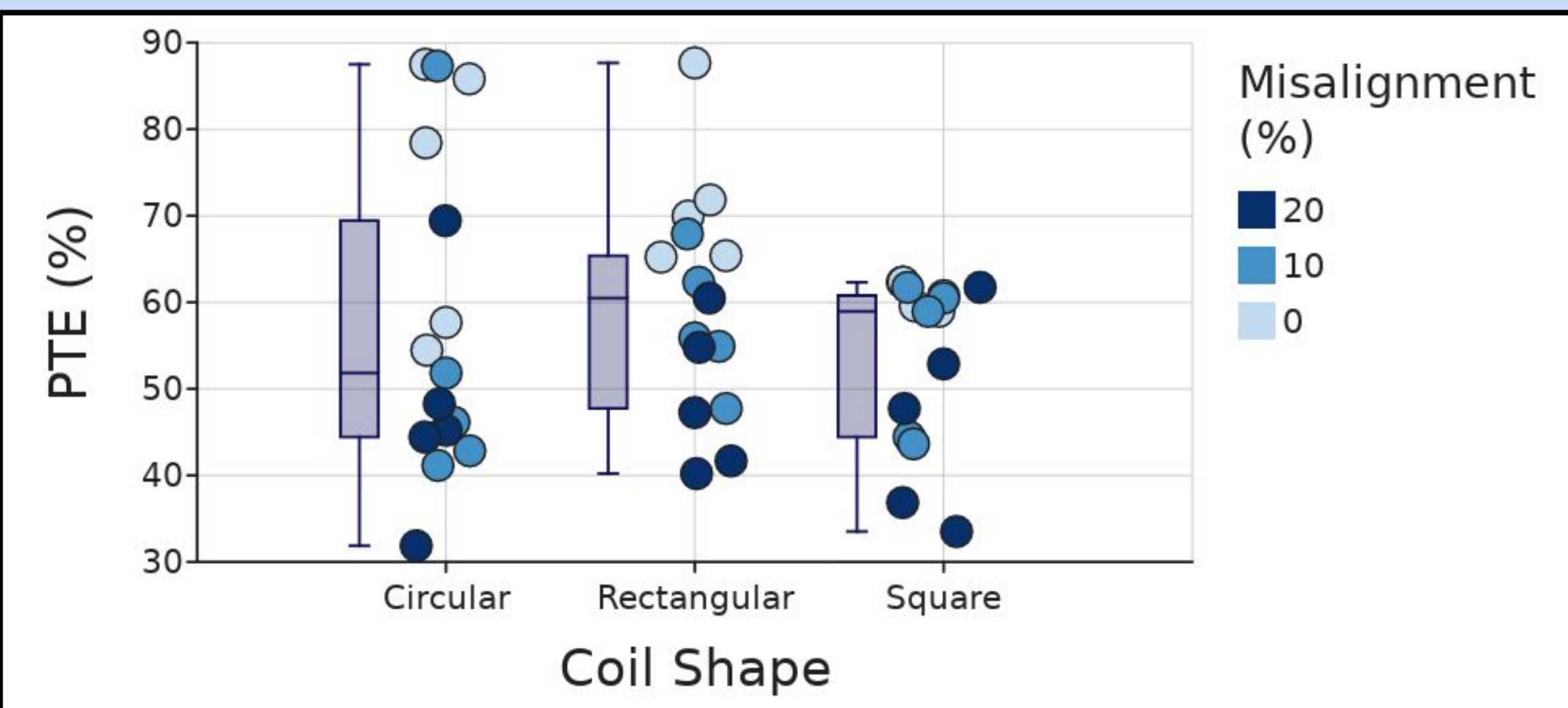


Figure 6: Box and whisker plot of PTE changes by coil shape. Image created by researcher using DataClassroom, 2026.

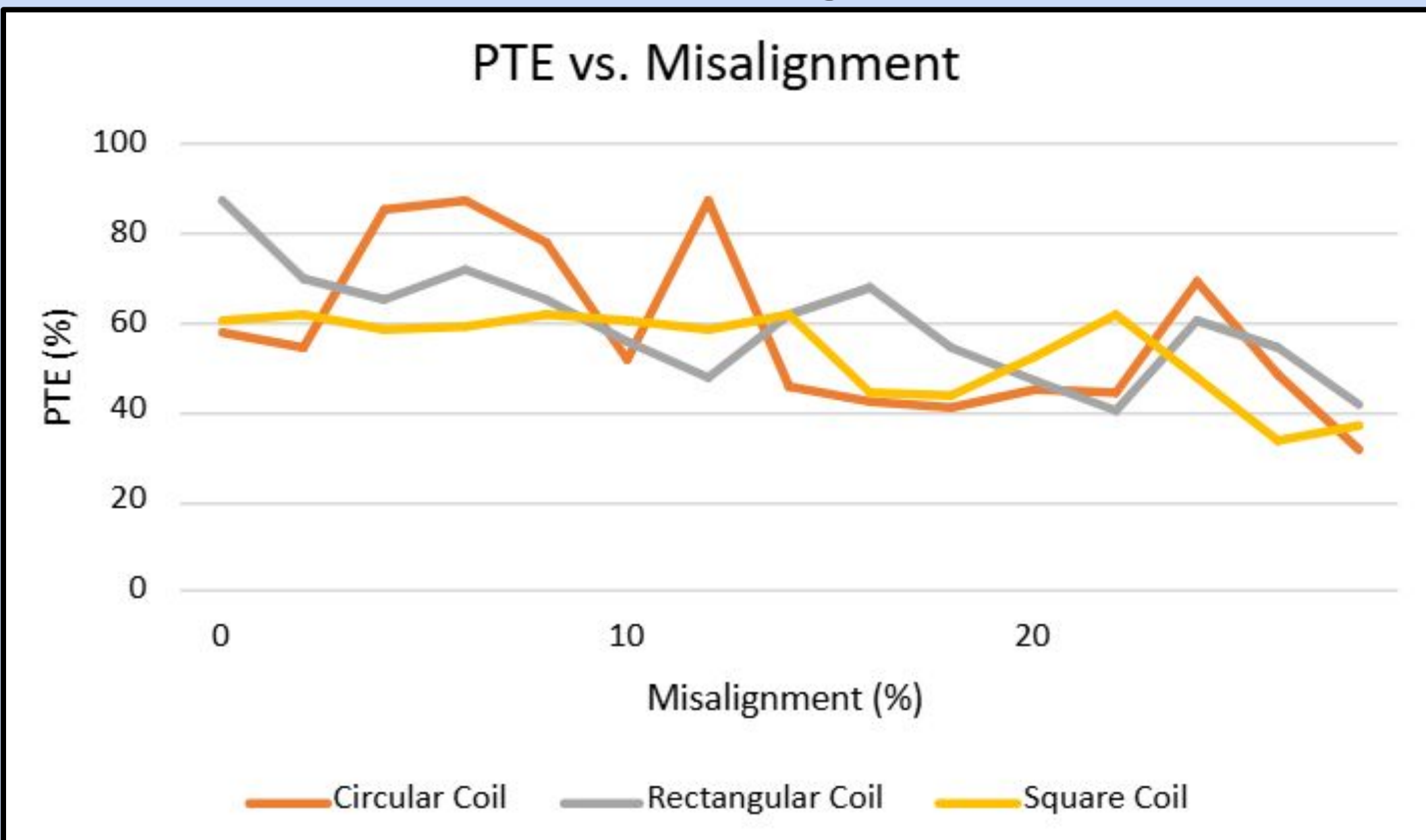


Figure 5: Graph of PTE changes by misalignment for all coil shapes. Image created by researcher using Excel, 2026.

- Figure 6 shows that rectangular coils achieved the **greatest median PTE**.
- Square coils had the **least variability** and **lowest peak PTE** between misalignment conditions.
- Circular coils had the **most data variability**.

Statistical Analysis

Pair	P-value
Circular - Rectangular	1.00
Circular - Square	1.00
Rectangular - Square	0.81

Figure 7: Post-hoc Dunn's Test for Kruskal-Wallis. Image created by researcher using DataClassroom, 2026.

Figure 7 shows the results of a **post-hoc Dunn's Test** comparing the PTE data from all three coil shapes to each other. The **high p-values of 1.00 and 0.81** fail to

reject the null hypothesis, suggesting that there is **no significant difference in PTE between coil shapes**.

Conclusions

Statistical Conclusion

- Coil geometry **did not significantly impact** overall performance.

Practical Conclusion

- The data **support existing research** showing that **coil misalignment decreases WRIPT efficiency**, especially in dynamic charging applications.
- However, coil shape **may not play as large a role** in shaping an efficient WRIPT system as **other prolific factors**, such as **air gap** or **shielding**.

Future Research

- Test more forms of **misalignment** (angular, air gap).
- Compare additional coil geometries.
- Create a **moving receiver system** to simulate dynamic WPT applications.

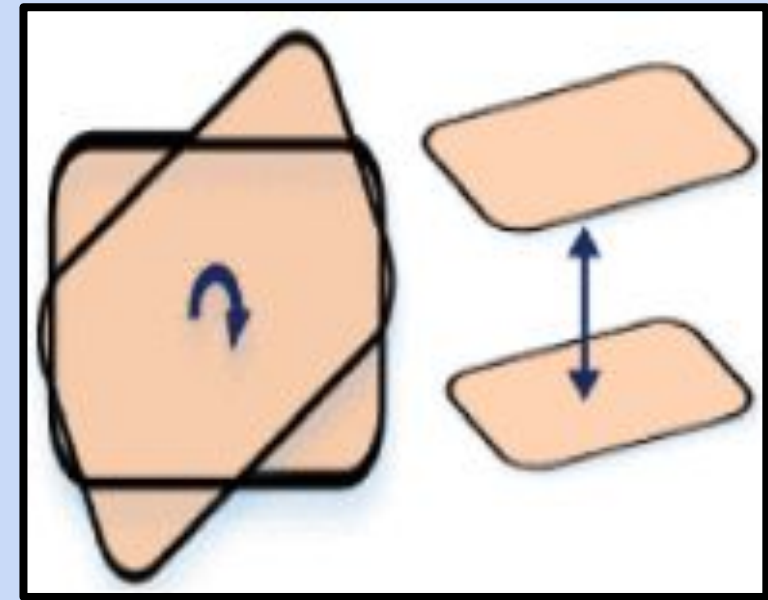


Figure 8: Illustration of angular (left) and air gap (right) misalignment. [1]

References

- [1] Rahul Kumar, J., Narayanamoorthi, R., Vishnuram, P., Bajaj, M., Blazek, V., Prokop, L., & Misak, S. (2022). An empirical survey on wireless inductive power pad and resonant magnetic field coupling for in-motion EV charging system. *IEEE Access*, 11. <https://doi.org/10.1109/ACCESS.2022.3232852>
- [2] Stout, T. (2020). *Experts talk: Wireless power transfer and electric vehicle applications with Thomas Stout | HDR*. [www.hdrinc.com](https://www.hdrinc.com/insights/experts-talk-wireless-power-transfer-and-electric-vehicle-applications-thomas-stout).
- [3] Wang, X., Pang, J., Zhao, N., Liu, L., Dong, H., Tan, Q., & Xiong, J. (2022). Lateral and angular misalignments of coil in wireless power transfer system. *Sensors and Actuators A: Physical*, 341. <https://doi.org/10.1016/j.sna.2022.113577>